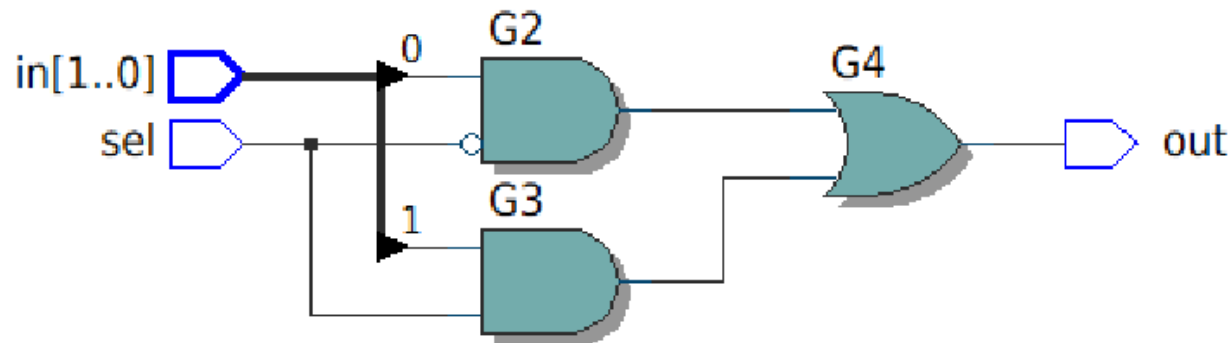


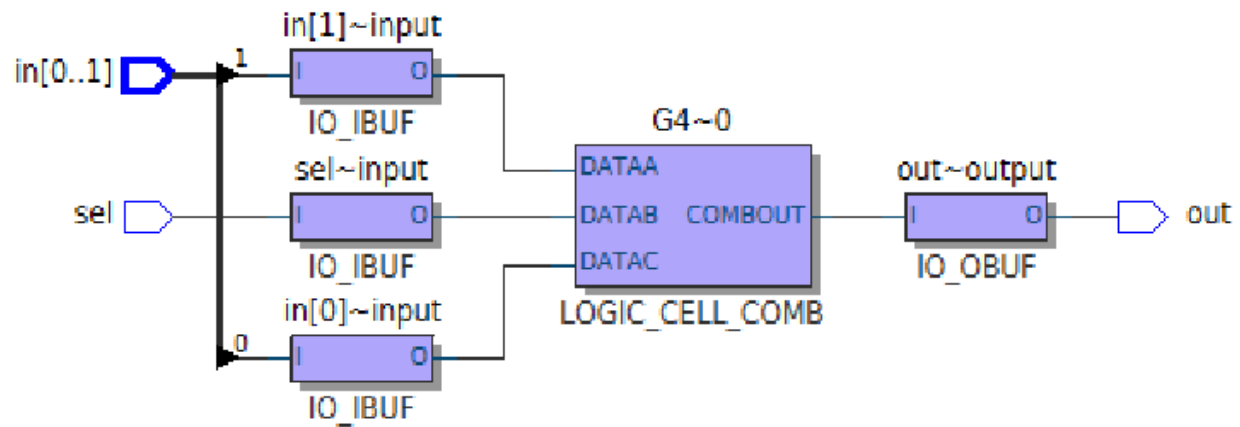
Mux 2x1

1. GATE

a. RTL Schematic:



b. Post Mapping Schematic:

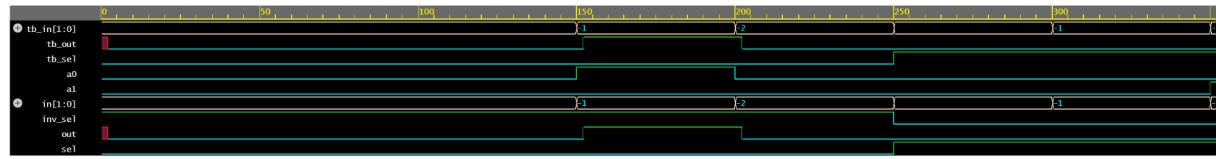


c. Resource Usage:

	Resource	Usage
1	Estimate of Logic utilization (ALMs needed)	1
2		
3	▼ Combinational ALUT usage for logic	1
1	-- 7 input functions	0
2	-- 6 input functions	0
3	-- 5 input functions	0
4	-- 4 input functions	0
5	-- ≤ 3 input functions	1
4		
5	Dedicated logic registers	0
6		
7	I/O pins	4
8		
9	Total DSP Blocks	0
10		
11	Maximum fan-out node	G4~0
12	Maximum fan-out	1
13	Total fan-out	8
14	Average fan-out	0.89

There were 1 ALUTs used because only one was needed. There were 3 input functions used in order to accommodate the two bit input and the 1 bit sel, making 3 inputs to the ALUT.

d. Simulation and transcript:

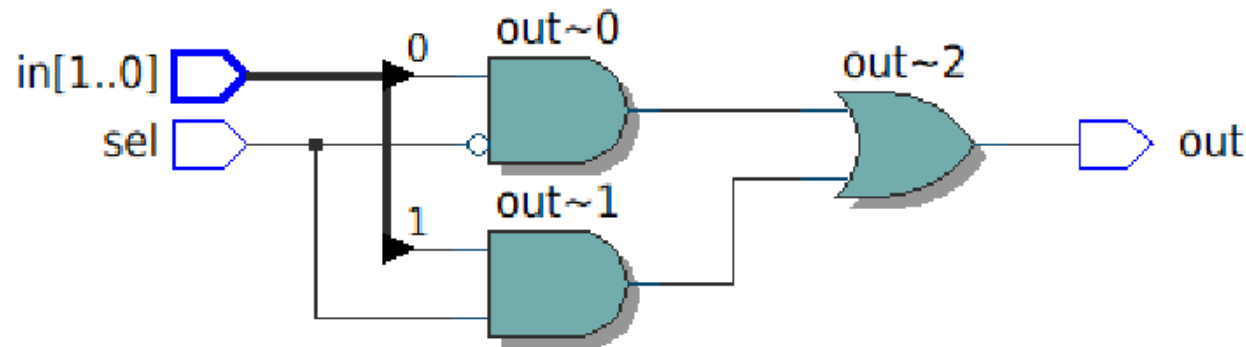


Note: To revert to EPWave opening in a new browser window, set that option on your profile page.

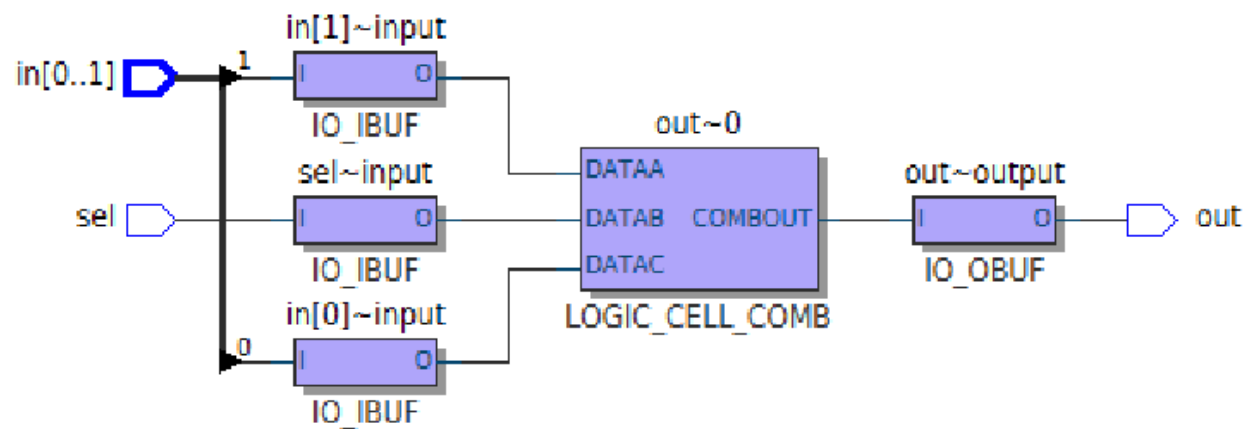
```
# Loading sv_std.std
# Loading work.mux_2x1_testbench(fast)
# Loading work.mux_2x1(fast)
#
# run -all
# time=0, in=00 sel=0 out=x
#
# time=2, in=00 sel=0 out=0
#
# time=150, in=01 sel=0 out=0
#
# time=152, in=01 sel=0 out=1
#
# time=200, in=10 sel=0 out=1
#
# time=202, in=10 sel=0 out=0
#
# time=250, in=00 sel=1 out=0
#
# time=300, in=01 sel=1 out=0
#
# time=350, in=10 sel=1 out=0
#
# time=352, in=10 sel=1 out=1
#
# exit
# End time: 18:34:25 on Jan 13,2026, Elapsed time: 0:00:01
# Errors: 0, Warnings: 0
End time: 18:34:25 on Jan 13,2026, Elapsed time: 0:00:02
*** Summary ***
    qrun: Errors: 0, Warnings: 0
    vlog: Errors: 0, Warnings: 0
    vopt: Errors: 0, Warnings: 1
    vsim: Errors: 0, Warnings: 0
    Totals: Errors: 0, Warnings: 1
Finding VCD file...
./dump.vcd
[2026-01-13 23:34:25 UTC] opening EPWave...
Done
```

2. DATAFLOW

a. RTL Schematic:



b. Post Mapping Schematic:

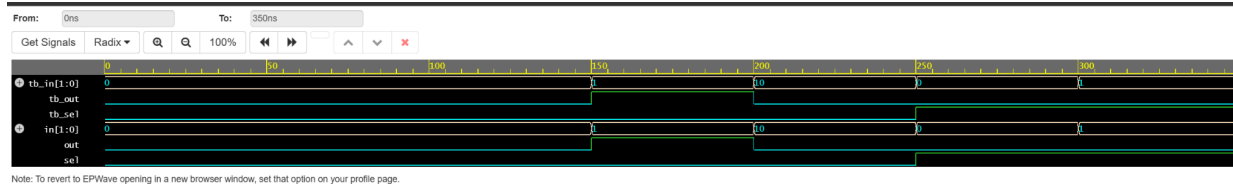


c. Resource Usage:

	Resource	Usage
1	Estimate of Logic utilization (ALMs needed)	1
2		
3	▼ Combinational ALUT usage for logic	1
1	-- 7 input functions	0
2	-- 6 input functions	0
3	-- 5 input functions	0
4	-- 4 input functions	0
5	-- ≤ 3 input functions	1
4		
5	Dedicated logic registers	0
6		
7	I/O pins	4
8		
9	Total DSP Blocks	0
10		
11	Maximum fan-out node	out~0
12	Maximum fan-out	1
13	Total fan-out	8
14	Average fan-out	0.89

Similar to the last one, there is 1 ALUT for three functions, as the process is essentially the same as described above

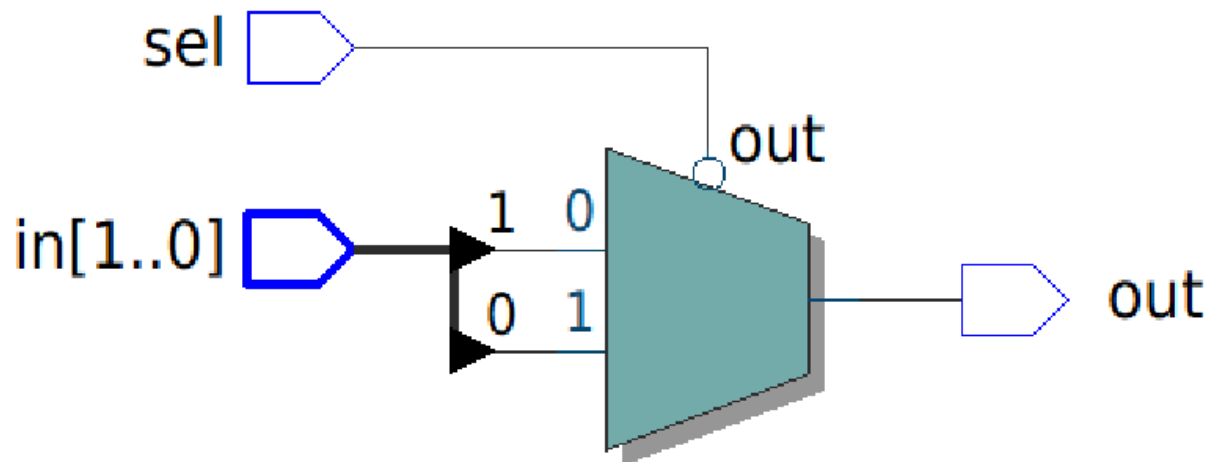
d. Simulation and transcript:



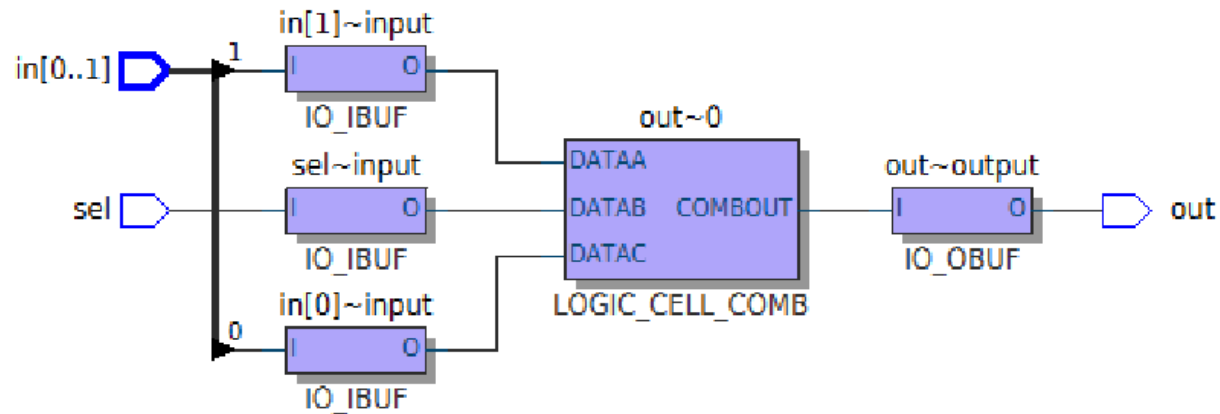
```
# Loading work.mux_2x1(fast)
#
# run -all
# time=0, in=00 sel=0 out=0
#
# time=150, in=01 sel=0 out=1
#
# time=200, in=10 sel=0 out=0
#
# time=250, in=00 sel=1 out=0
#
# time=300, in=01 sel=1 out=0
#
# time=350, in=10 sel=1 out=1
#
# exit
# End time: 14:19:44 on Jan 15,2026, Elapsed time: 0:00:01
```

3. BEHAVIORAL

a. RTL Schematic:



b. Post Mapping Schematic:

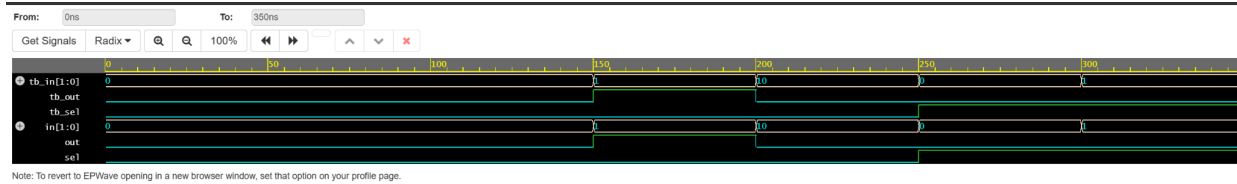


c. Resource Usage:

	Resource	Usage
1	Estimate of Logic utilization (ALMs needed)	1
2		
3	▼ Combinational ALUT usage for logic	1
1	-- 7 input functions	0
2	-- 6 input functions	0
3	-- 5 input functions	0
4	-- 4 input functions	0
5	-- <=3 input functions	1
4		
5	Dedicated logic registers	0
6		
7	I/O pins	4
8		
9	Total DSP Blocks	0
10		
11	Maximum fan-out node	out~0
12	Maximum fan-out	1
13	Total fan-out	8
14	Average fan-out	0.89

(see above for explanation)

d. Simulation and transcript:

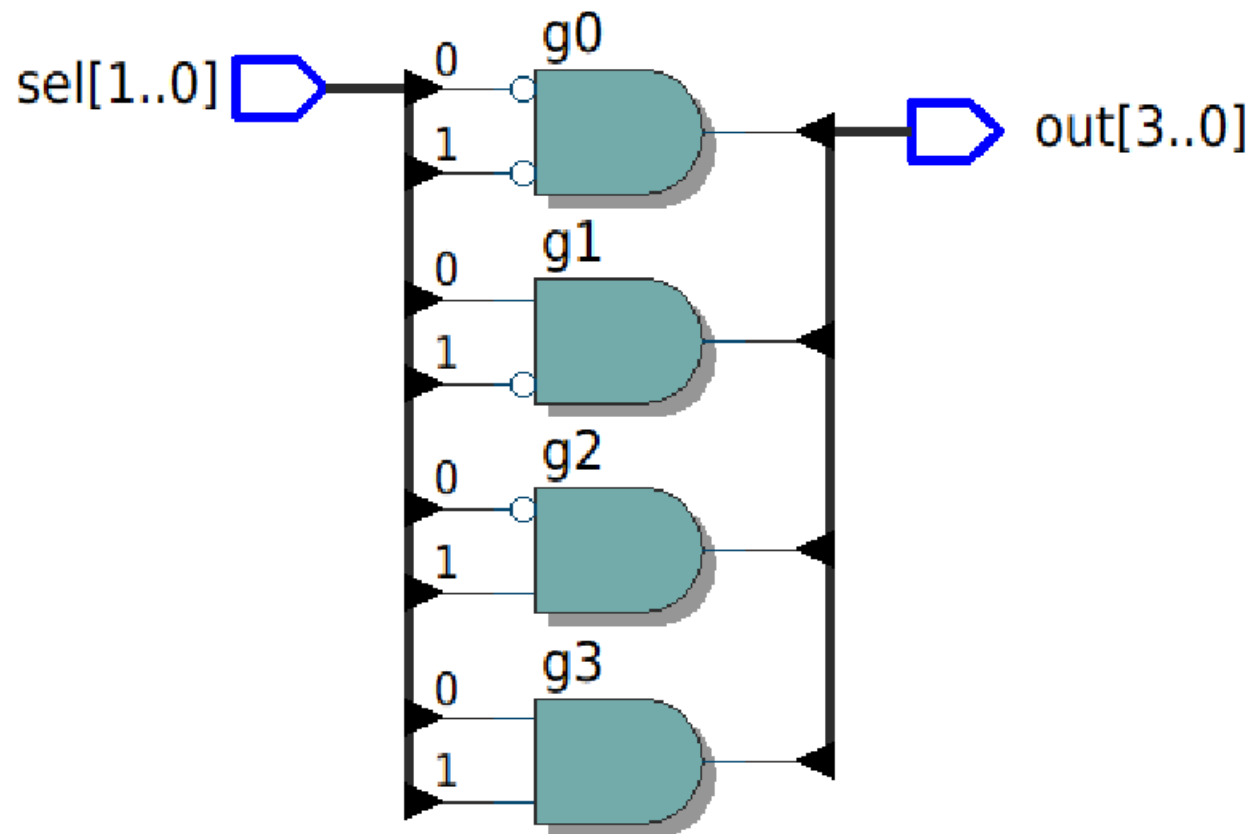


```
# Loading work.mux_2x1(fast)
#
# run -all
# time=0, in=00 sel=0 out=0
#
# time=150, in=01 sel=0 out=1
#
# time=200, in=10 sel=0 out=0
#
# time=250, in=00 sel=1 out=0
#
# time=300, in=01 sel=1 out=0
#
# time=350, in=10 sel=1 out=1
#
# exit
# End time: 14:18:49 on Jan 15,2026, Elapsed time: 0:00:01
```

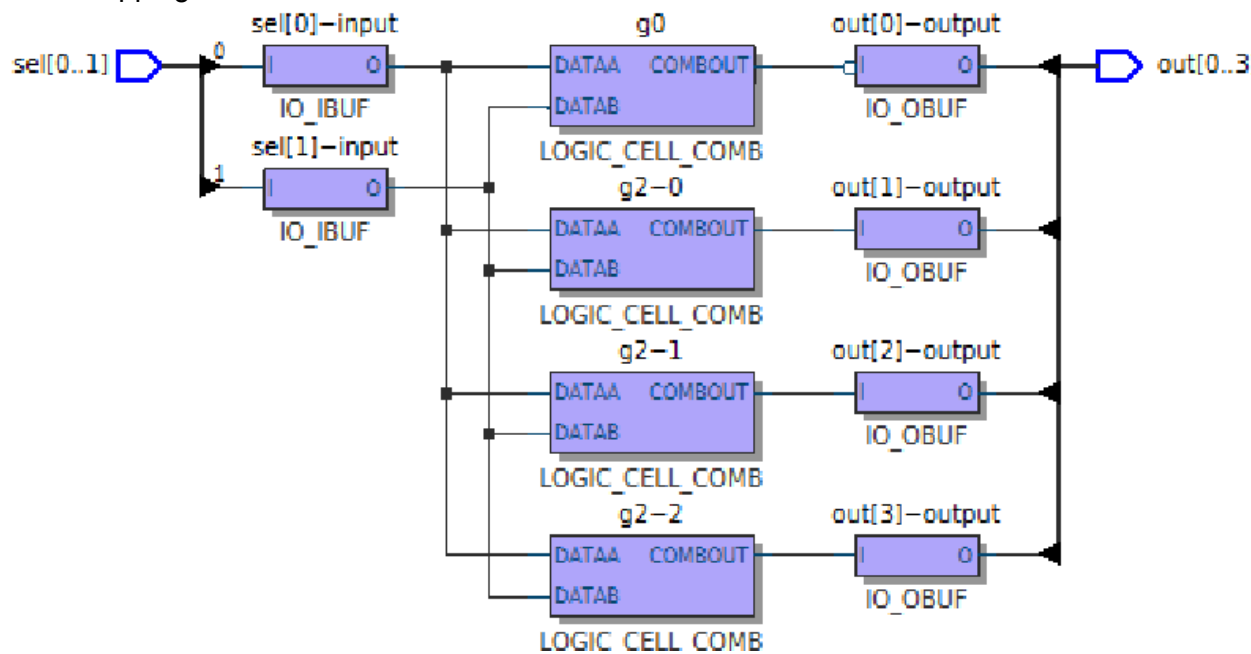
Decoder

1. GATE

a. RTL



b. Post Mapping Schematic:

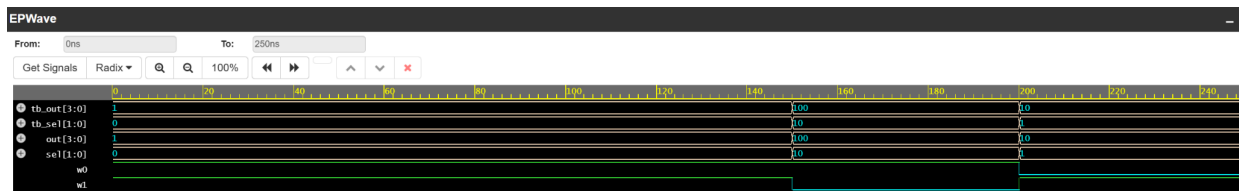


c. Resource Usage:

	Resource	Usage
1	Estimate of Logic utilization (ALMs needed)	2
2		
3	▼ Combinational ALUT usage for logic	4
1	-- 7 input functions	0
2	-- 6 input functions	0
3	-- 5 input functions	0
4	-- 4 input functions	0
5	-- ≤ 3 input functions	4
4		
5	Dedicated logic registers	0
6		
7	I/O pins	6
8		
9	Total DSP Blocks	0
10		
11	Maximum fan-out node	sel...put
12	Maximum fan-out	4
13	Total fan-out	18
14	Average fan-out	1.12

The number of ALUTs used was 4 because that's the number of logical decisions made based on a combination of the input bits in order to decode the output string, all with ≤ 3 input functions because there are only 2 input bits

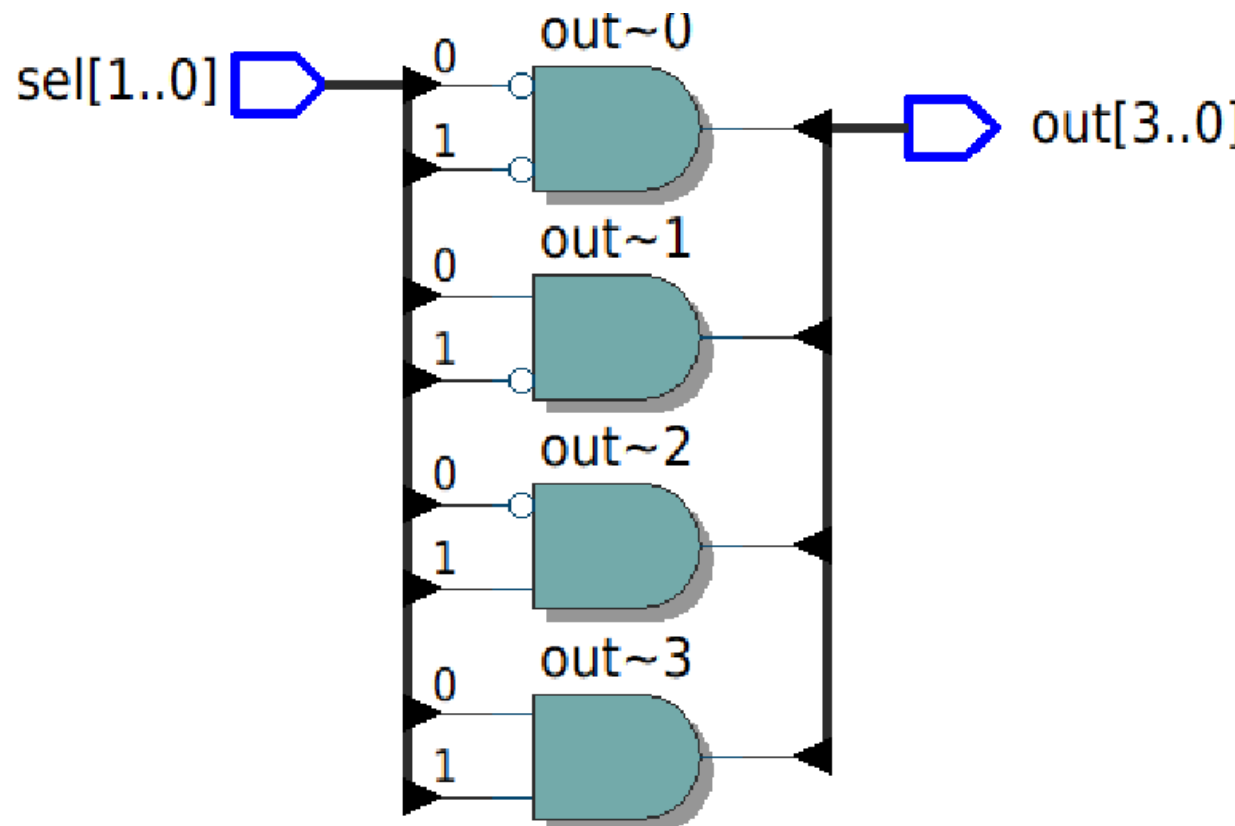
d. Simulation and Transcript:



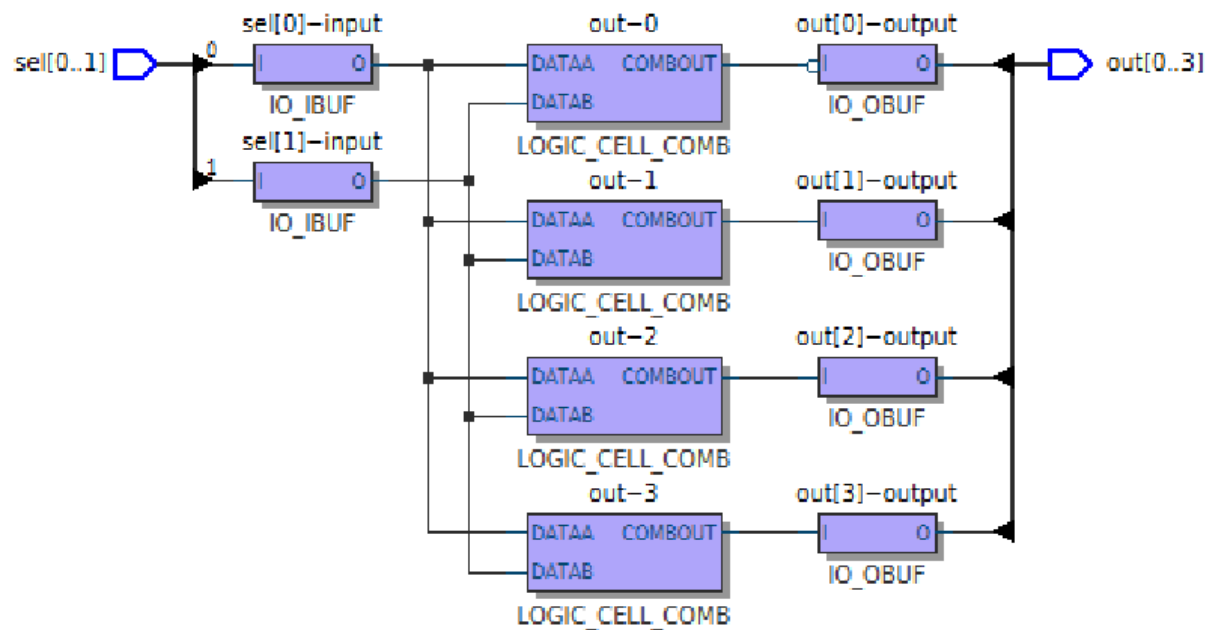
```
# Loading work.decoder_2to4_testbench(fast)
# Loading work.decoder(fast)
#
# run -all
# time=0, sel=00 out=0001
#
# time=150, sel=10 out=0100
#
# time=200, sel=01 out=0010
#
# time=250, sel=11 out=1000
#
# exit
# End time: 14:16:52 on Jan 15,2026, Elapsed time: 0:00:01
# Errors: 0, Warnings: 0
```

2. DATAFLOW

a. RTL Schematic:



b. Post Mapping Schematic:

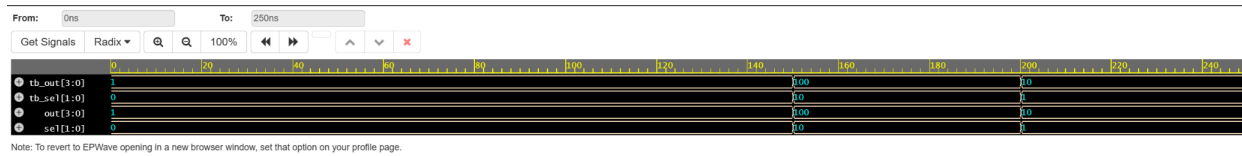


c. Resource Usage:

	Resource	Usage
1	Estimate of Logic utilization (ALMs needed)	2
2		
3	▼ Combinational ALUT usage for logic	4
1	-- 7 input functions	0
2	-- 6 input functions	0
3	-- 5 input functions	0
4	-- 4 input functions	0
5	-- <=3 input functions	4
4		
5	Dedicated logic registers	0
6		
7	I/O pins	6
8		
9	Total DSP Blocks	0
10		
11	Maximum fan-out node	sel...put
12	Maximum fan-out	4
13	Total fan-out	18
14	Average fan-out	1.12

As we can tell by the above diagrams and usage summary, these implementations are very similar so the explanations for resource usage apply from the first answer.

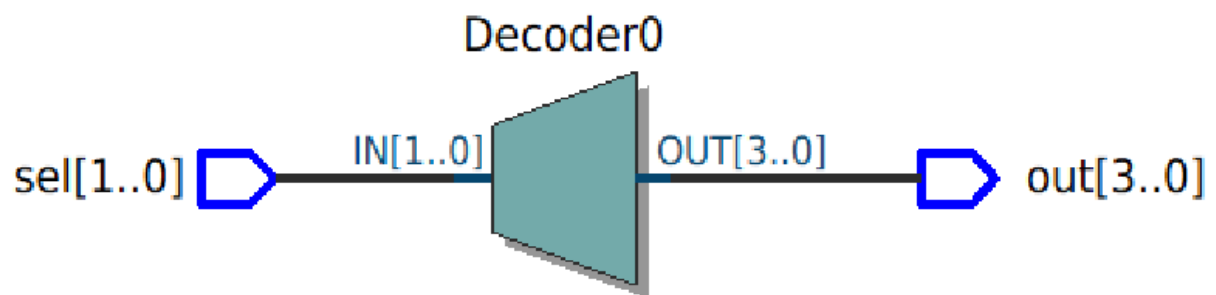
d. Simulation and transcript::



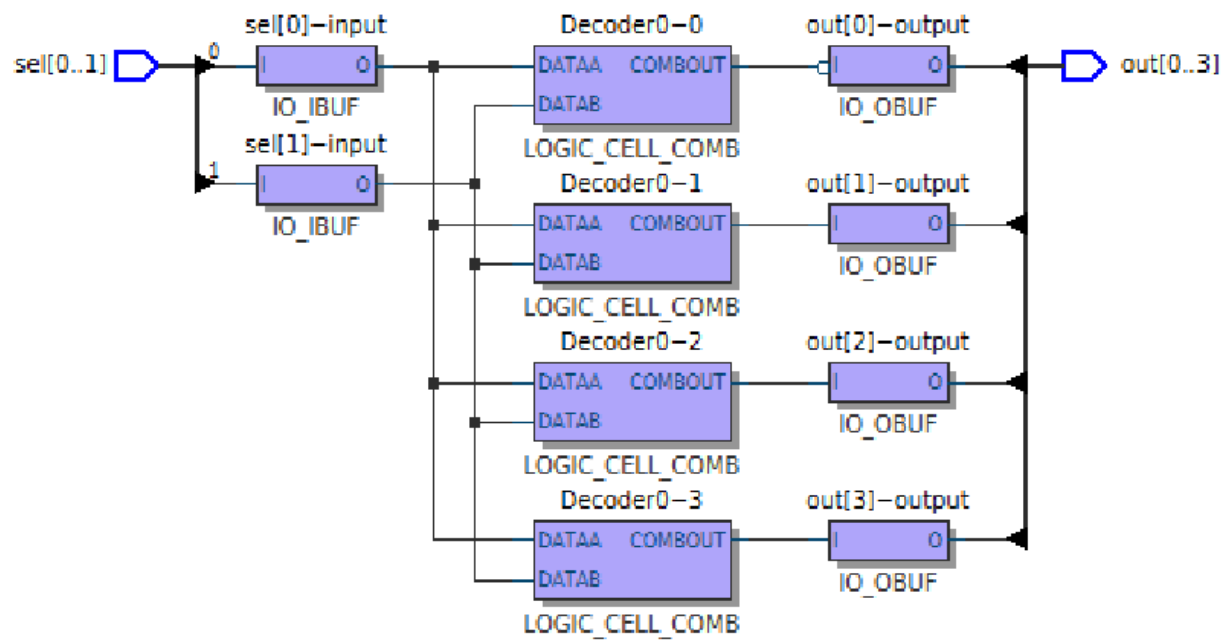
```
// Loading work.decoder_fast,
# Loading work.decoder(fast)
#
# run -all
# time=0, sel=00 out=0001
#
# time=150, sel=10 out=0100
#
# time=200, sel=01 out=0010
#
# time=250, sel=11 out=1000
#
# exit
```

3. BEHAVIORAL

a. RTL Schematic:



b. Post Mapping Schematic:

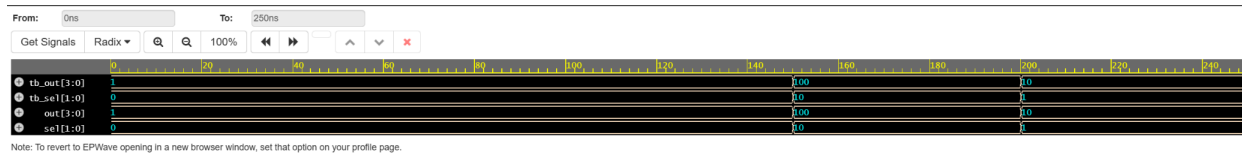


c. Resource Usage:

	Resource	Usage
1	Estimate of Logic utilization (ALMs needed)	2
2		
3	▼ Combinational ALUT usage for logic	4
1	-- 7 input functions	0
2	-- 6 input functions	0
3	-- 5 input functions	0
4	-- 4 input functions	0
5	-- ≤ 3 input functions	4
4		
5	Dedicated logic registers	0
6		
7	I/O pins	6
8		
9	Total DSP Blocks	0
10		
11	Maximum fan-out node	sel...put
12	Maximum fan-out	4
13	Total fan-out	18
14	Average fan-out	1.12

The usage summary can be explained by noticing the post-mapping is the same, regardless of method used to achieve the function.

d. Simulation and transcript:

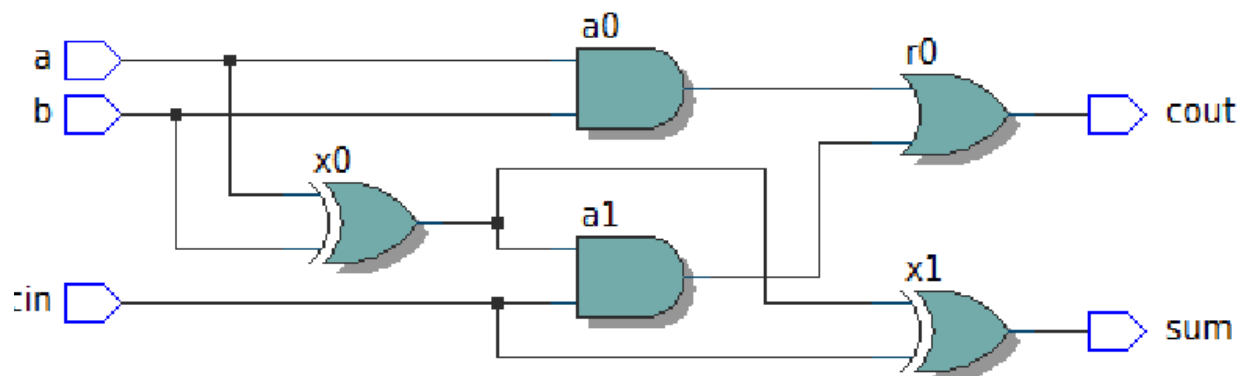


```
" Loading work.decoder(fast)
#
# run -all
# time=0, sel=00 out=0001
#
# time=150, sel=10 out=0100
#
# time=200, sel=01 out=0010
#
# time=250, sel=11 out=1000
#
# exit
```

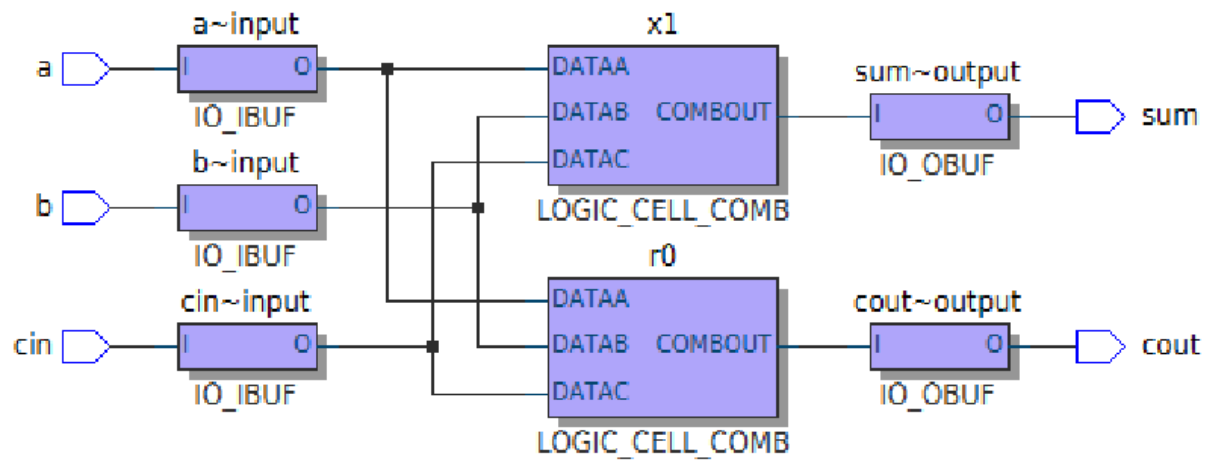
Full Adder

1. GATE

a. RTL Schematic:



b. Post Mapping Schematic:



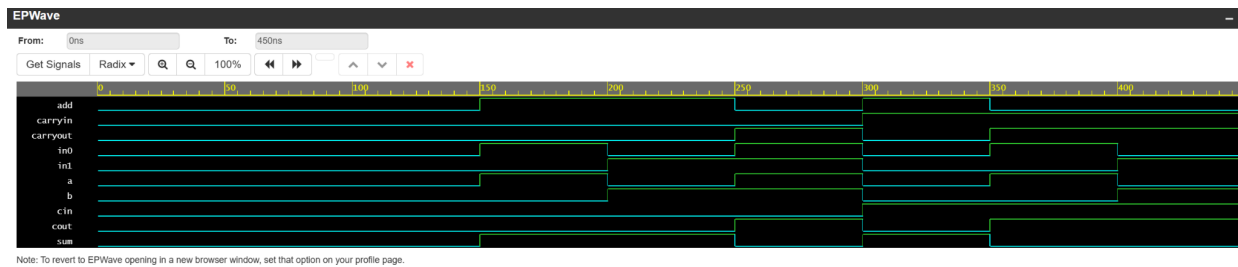
c. Resource Usage:

	Resource	Usage
1	Estimate of Logic utilization (ALMs needed)	1
2		
3	▼ Combinational ALUT usage for logic	2
1	-- 7 input functions	0
2	-- 6 input functions	0
3	-- 5 input functions	0
4	-- 4 input functions	0
5	-- <=3 input functions	2
4		
5	Dedicated logic registers	0
6		
7	I/O pins	5
8		
9	Total DS I/O pins	0
10		
11	Maximum fan-out node	a~input
12	Maximum fan-out	2
13	Total fan-out	13
14	Average fan-out	1.08

There were 2 ALUTs used for the full adder. These ALUTs took 3 inputs and

outputted 1 output. The logic can be seen as based on the combination of two input bits and the carry-in bit, these ALUTs determine the sum bit and carry-out bit.

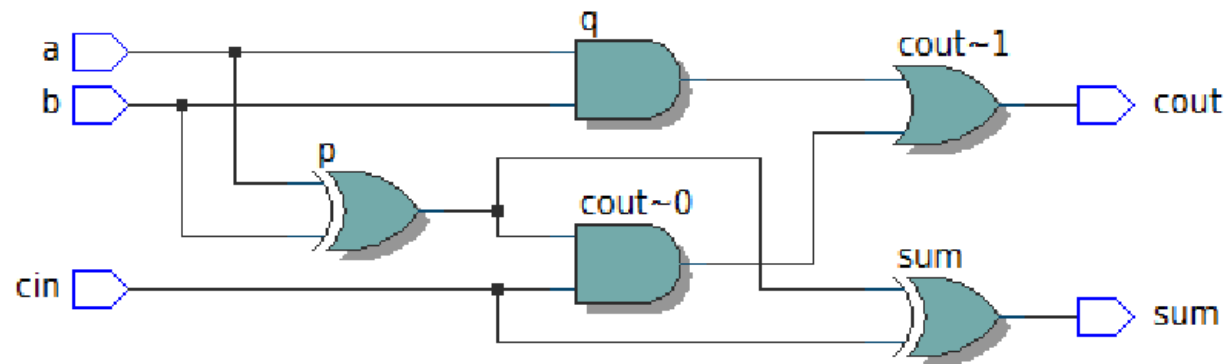
d. Simulation and transcript:v



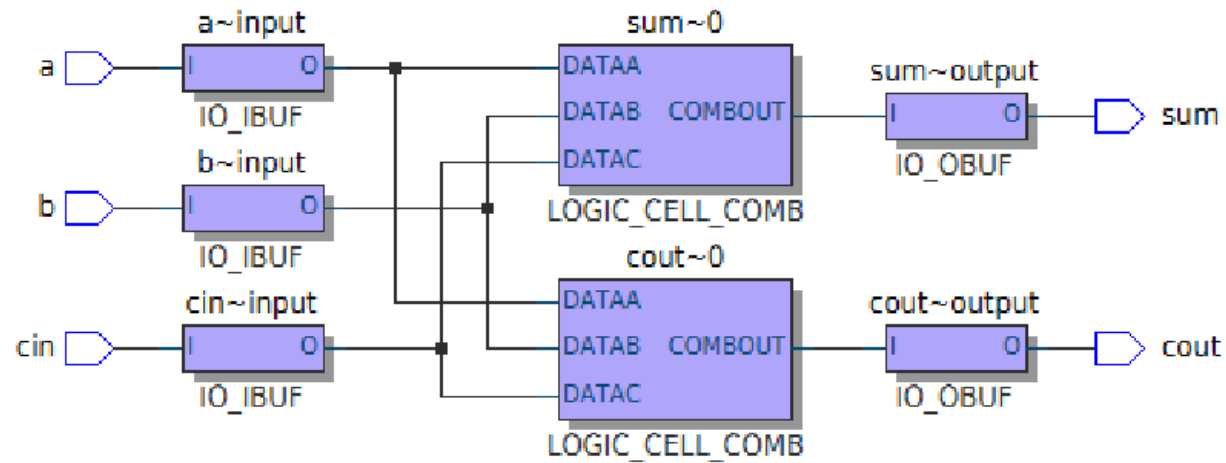
```
# run -all
# time=0    a=0    b=0    c=0    sum=0    cout=0
#
# time=150   a=1    b=0    c=0    sum=1    cout=0
#
# time=200   a=0    b=1    c=0    sum=1    cout=0
#
# time=250   a=1    b=1    c=0    sum=0    cout=1
#
# time=300   a=0    b=0    c=1    sum=1    cout=0
#
# time=350   a=1    b=0    c=1    sum=0    cout=1
#
# time=400   a=0    b=1    c=1    sum=0    cout=1
#
# time=450   a=1    b=1    c=1    sum=1    cout=1
#
# exit
```

2. DATAFLOW

a. RTL Schematic:



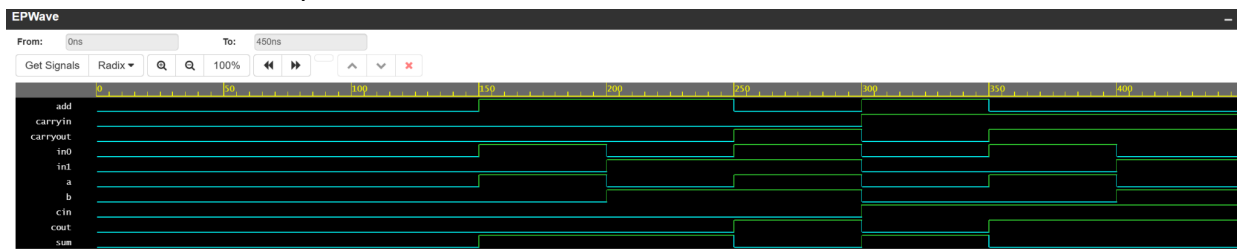
b. Post Mapping Schematic:



c. Resource Usage:

	Resource	Usage
1	Estimate of Logic utilization (ALMs needed)	1
2		
3	▼ Combinational ALUT usage for logic	2
1	-- 7 input functions	0
2	-- 6 input functions	0
3	-- 5 input functions	0
4	-- 4 input functions	0
5	-- ≤ 3 input functions	2
4		
5	Dedicated logic registers	0
6		
7	I/O pins	5
8		
9	Total DSP Blocks	0
10		
11	Maximum fan-out node	a~input
12	Maximum fan-out	2
13	Total fan-out	13
14	Average fan-out	1.08

d. Simulation and transcript:

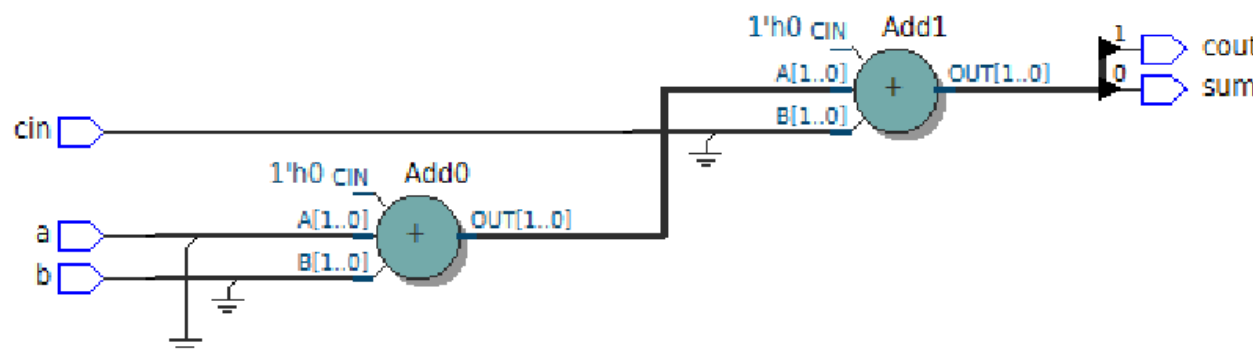


Note: To revert to EPWave opening in a new browser window, set that option on your profile page.

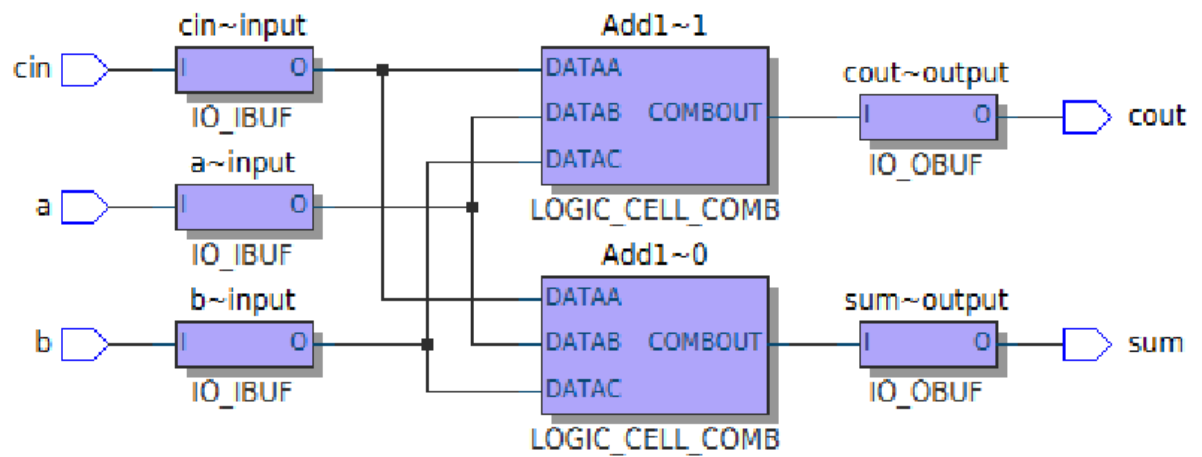
```
# run -all
# time=0    a=0    b=0    c=0    sum=0    cout=0
#
# time=150   a=1    b=0    c=0    sum=1    cout=0
#
# time=200   a=0    b=1    c=0    sum=1    cout=0
#
# time=250   a=1    b=1    c=0    sum=0    cout=1
#
# time=300   a=0    b=0    c=1    sum=1    cout=0
#
# time=350   a=1    b=0    c=1    sum=0    cout=1
#
# time=400   a=0    b=1    c=1    sum=0    cout=1
#
# time=450   a=1    b=1    c=1    sum=1    cout=1
#
# exit
```

3. BEHAVIORAL

a. RTL Schematic:



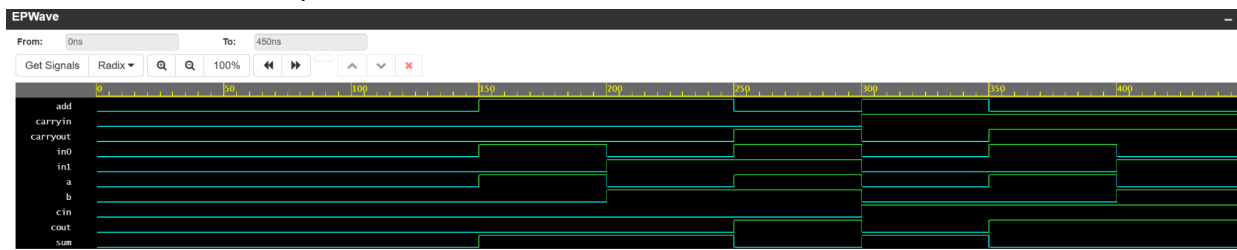
b. Post Mapping Schematic:



c. Resource Usage:

	Resource	Usage
1	Estimate of Logic utilization (ALMs needed)	1
2		
3	▼ Combinational ALUT usage for logic	2
1	-- 7 input functions	0
2	-- 6 input functions	0
3	-- 5 input functions	0
4	-- 4 input functions	0
5	-- ≤ 3 input functions	2
4		
5	Dedicated logic registers	0
6		
7	I/O pins	5
8		
9	Total DSP Blocks	0
10		
11	Maximum fan-out node	cin...put
12	Maximum fan-out	2
13	Total fan-out	13
14	Average fan-out	1.08

d. Simulation and transcript:



Note: To revert to EPWave opening in a new browser window, set that option on your profile page.

```
# Loading work.fulladder(fast)
#
# run -all
# time=0    a=0    b=0    c=0    sum=0    cout=0
#
# time=150   a=1    b=0    c=0    sum=1    cout=0
#
# time=200   a=0    b=1    c=0    sum=1    cout=0
#
# time=250   a=1    b=1    c=0    sum=0    cout=1
#
# time=300   a=0    b=0    c=1    sum=1    cout=0
#
# time=350   a=1    b=0    c=1    sum=0    cout=1
#
# time=400   a=0    b=1    c=1    sum=0    cout=1
#
# time=450   a=1    b=1    c=1    sum=1    cout=1
#
# exit
```